An Introduction to CLEAR for Multimodal Healthcare Analytics

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ABSTRACT

The field of healthcare analytics uses electronic health records (EHRs) as one of its main sources of data for machine learning. The multimodal and heterogeneous nature of EHRs introduces the problem of representation contamination. This issue can be described as the entanglement of spurious or weakly informative data with genuinely discriminative features, where false correlations cause models to rely on irrelevant or misleading patterns. To address this issue, Su et al. (2025) propose CLEAR, a Counterfactual Disparity Learning Model for multimodal EHR analysis. [15]

CLEAR first models representation contamination by assessing the discrepancy between factual and counterfactual representations through the Counterfactual Prompt Learning (CPL) module. Thereafter, the Adaptive Dynamic Imputation (ADI) module adjusts these representations by suppressing low-discriminative features and amplifying informative ones. Lastly, the Multimodal Representation Fusion (MRF) module models intra- and inter-modality correlations via Multi-Head Self-Attention, achieving a truly multimodal representation.

The authors evaluated CLEAR on two real-world EHR datasets: MIMIC-IV and ODIR. The proposed approach consistently outperforms state-of-the-art models. These results suggest that explicitly modeling representation contamination is a promising direction for robust multimodal healthcare analytics.

CCS CONCEPTS

• **Applied computing** → **Health informatics**.

KEYWORDS

Healthcare Analytics, Electronic Health Records, Multimodal Learning, Deep Learning, Representation Learning

1 INTRODUCTION

Over the recent years, the adoption of electronic health records (EHRs) has steadily increased, both in the United States and internationally [8, 16]. Since the primary purpose of EHRs revolves around data exchange, billing, and education, their use in research is secondary, as the data is inherently heterogeneous and noisy. Machine learning offers promising methods to address these challenges, as techniques such as natural language processing and image recognition are capable of extracting meaningful insights from the unstructured data inherent to EHRs [8]. In particular, the inclusion of diverse data modalities such as demographic information, medical imaging, time-series data, and clinical text records offers significant potential for healthcare analytics. Recent advances in multimodal learning have demonstrated clear advantages over unimodal approaches. [15]. Models such as RAIM [21], which combines continuous monitoring data with discrete clinical events, and FUSION [22], which integrates clinical notes with multivariate time series, illustrate the potential of fusing heterogeneous EHR modalities. However, these methods largely overlook a critical challenge inherent to multimodal EHR data: representation contamination.

2 PROBLEM STATEMENT

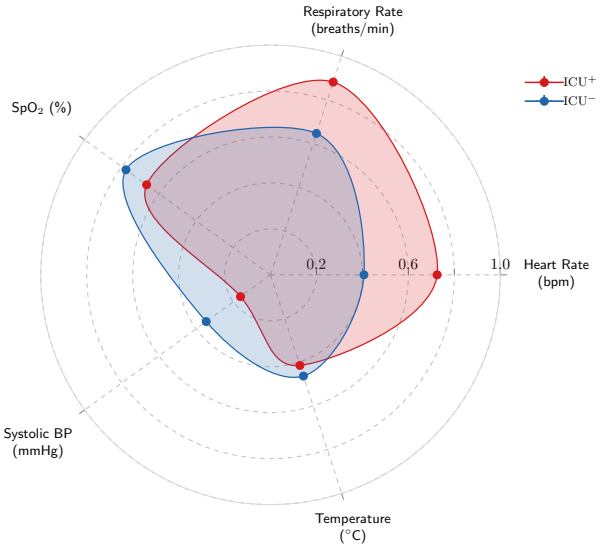
In order to recognize the core problem that CLEAR addresses, it is essential to understand the peculiarities of discriminative and spurious features. Discriminative features refer to physiological patterns that are directly associated with true patient deterioration, typically reflected in high heart rate and respiratory rate, and low blood pressure and oxygen saturation, as used in early warning systems. In contrast, spurious features are correlations that appear predictive within the data but do not reflect the underlying pathophysiological deterioration process. [3, 15]

As an illustrative example, we consider two representative patients: one who experienced clinical deterioration requiring ICU transfer within 6 hours (positive case) and one who did not (negative case). As shown in Figure 1a, the positive case (ICU^+) exhibits clear discriminative features, including increased heart rate and respiratory rate combined with moderately reduced blood pressure and oxygen saturation, consistent with established physiological deterioration patterns. In comparison, the negative case (ICU^-) acts as a baseline illustrating stable vital parameters.

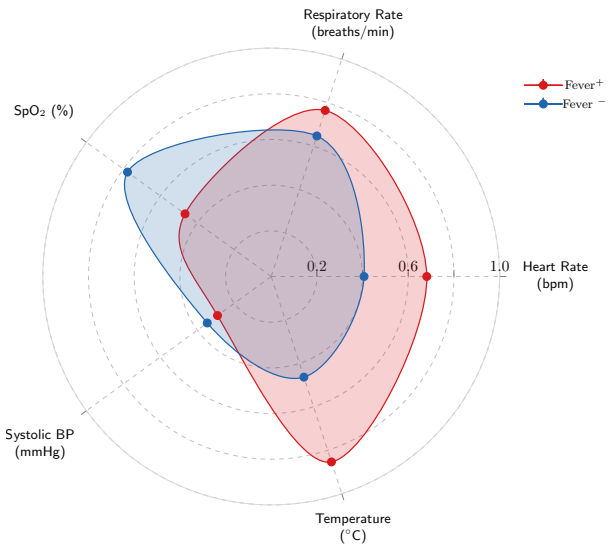
In order to demonstrate a spurious representation, we further focus on the feature "temperature". In contrast to other vital signs, it shows little to no variation with respect to the prediction target "ICU transfer within 6 hours," (cf. Figure 1a) indicating that it is a less discriminative feature for this task. The problem arises when reviewing the Fever related prompt. In Figure 1b, a febrile patient ($Fever^+$) exhibits similar vital signs to the positive patient (ICU^+), with the exception of an elevated body temperature consistent with fever. This leads models to associate elevated temperature with the ICU transfer prompt, despite its weak direct relation to imminent deterioration, resulting in potential representation contamination. Consequently, the model may incorrectly amplify temperature-related patterns during training and rely on clinically irrelevant correlations, thereby reducing its ability to distinguish truly deteriorating patients from physiologically stable febrile cases. [3, 10]

3 CLEAR: METHOD OVERVIEW

In this section, we introduce the underlying methodology of CLEAR and describe its three core modules: Counterfactual Prompt Learning (CPL), Adaptive Dynamic Imputation (ADI), and Multimodal Representation Fusion (MRF). As discussed in the previous section, multimodal EHR data contains both discriminative and spurious correlations, which can contaminate latent representations during multimodal fusion. Conventional multimodal healthcare



(a) ICU transfer prediction (ICU$^+$ vs. ICU$^-$).



(b) Fever classification (Fever$^+$ vs. Fever$^-$).

Figure 1: Radar plots comparing normalized vital signs across ICU transfer and fever classification tasks.

models mainly focus on modality-specific feature extraction and representation fusion while overlooking the issue of representation contamination [15, 21, 22].

CLEAR addresses this limitation by explicitly modeling and refining contaminated representations. Figure 2 illustrates the overall pipeline of the framework. First, modality-specific encoders extract latent representations from heterogeneous EHR modalities such as structured clinical variables, time-series measurements, and clinical notes. Next, the CPL module captures representation discrepancies between factual and counterfactual prompts via the discrepancy function CRD. Based on these discrepancies, the ADI module suppresses weakly informative features and enhances clinically relevant ones. Concerning our ICU prediction example the ICU$^+$ condition is driven by concurrent abnormalities in heart rate, respiratory rate, oxygen saturation, and blood pressure, whereas body temperature remains largely invariant across factual and counterfactual settings, contributing little to their separation (cf. Figure 1a). Consequently, ADI downweights temperature in the final representation. Finally, the MRF module models intra- and inter-modality interactions to construct a unified multimodal representation for downstream prediction tasks.

In the following we will take a closer look on the underlying functionality and understand how each component contributes to counterfactual separability and multimodal consistency in detail.

3.1 Modality-specific Representation Modeling

Modality-specific representation modeling constitutes the initial stage of the CLEAR architecture. Recent studies have shown that it represents the de facto standard for uniformly modeling heterogeneous data [9, 23, 25]. We denote x_i^m as the raw multimodal input of modality m for patient i. The corresponding modality-specific representation r_i^m is extracted using a feature extractor. CLEAR utilizes different neural network architectures tailored to each data type: a Multi-Layer Perceptron (MLP) for discrete variables [13], a bidirectional GRU for time-series data [14], ResNet for image-based inputs [5], and ClinicalBERT for textual information [18]. Thus, MRM's representations r_i^m remain prone to latent contamination, requiring further refinement in CLEAR.

3.2 Counterfactual Prompt Learning

Building upon the MRM framework described above, we obtain modality-specific representations r_i^m for each patient and modality. The Counterfactual Prompt Learning (CPL) module operates on these representations by constructing factual and counterfactual prompts to expose discriminative representation discrepancies. For each patient, a factual prompt is derived from the ground-truth label (e.g., "The patient requires ICU transfer within 6 hours"), while a counterfactual prompt negates this outcome. This design is inspired by PromptEM [19], enabling the model to construct semantically contrasting prompts that highlight class-specific differences in the latent representation space. Both prompts are encoded using ClinicalBERT and used to project each modality-specific representation into factual and counterfactual task-aware spaces. The resulting counterfactual representation discrepancy (CRD), defined as the element-wise subtraction between factual and counterfactual representations, captures the degree to which each feature contributes to separating the two conditions [15].

3.3 Adaptive Dynamic Imputation

Building upon the counterfactual representation discrepancies captured by the CPL module, the Adaptive Dynamic Imputation (ADI) module performs element-wise calibration and enhancement. The

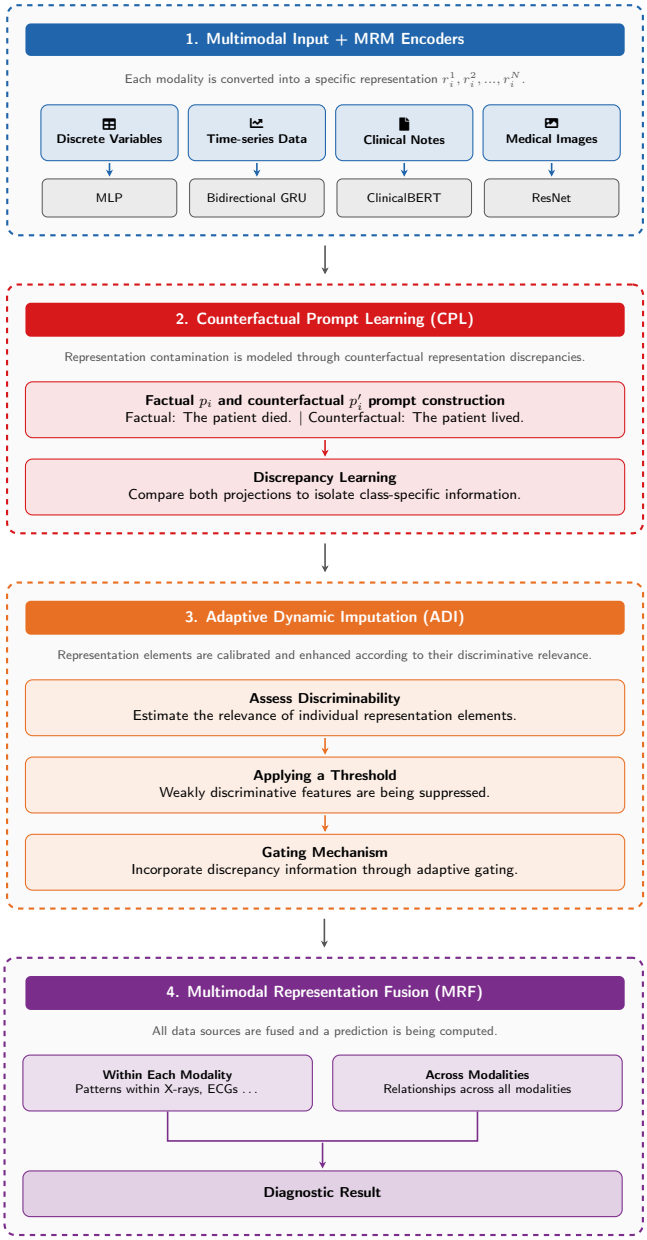


Figure 2: Overview of the CLEAR model architecture (adapted from Su et al. [15])

values obtained from the CRD (Counterfactual Representation Discrepancy) function are mapped to the interval $(0, 1)$ using the sigmoid activation function. Furthermore, a hyperparameter τ is employed as a threshold to suppress spurious features. More specifically, every representation whose discrepancy score falls below the threshold τ is set to 0 and is therefore excluded from further consideration by the model.

A gating mechanism concludes the ADI module by adaptively integrating the calibrated counterfactual discrepancy information into the original feature representations. For each feature dimension independently, the mechanism determines how strongly the counterfactually corrected representation should override the original one. Only those dimensions whose discrepancy scores exceed the threshold τ are retained, while all remaining dimensions are suppressed. Consequently, only highly discriminative information is incorporated into the enhanced representations, preventing spurious or uninformative corrections from corrupting the learned features [15].

Table 1: Statistics of the MIMIC-IV and ODIR Datasets

Dataset	MIMIC-IV	ODIR
# samples	418,100	5,000
# modalities	5	3
# modality features	(65, 7, 704, 10320, 768)	(2, 2·768, 2·1000)
# tasks	2	1
% positive labels	[47.76, 6.63]	32.48
% train:val:test split	8:1:1	7:1:2

3.4 Multimodal Representation Fusion

The Multimodal Representation Fusion (MRF) module integrates the enhanced modality-specific representations across all modalities into a unified, highly discriminative representation. To this end, both intra-modality and inter-modality dynamic correlations are explicitly modeled.

First, intra-modality dependencies are explored separately for each modality using a stack of Multi-Head Self-Attention (MHSA) layers [17]. This allows the model to capture contextual relationships within the enhanced representation space while preserving modality-specific characteristics. The resulting representations are then concatenated and passed through a second stack of MHSA layers, which attends across modality boundaries to capture inter-modality dependencies. A subsequent feed-forward network maps the fused representation to the prediction space [15].

4 RESULTS

In order to assess the effectiveness of CLEAR, Su et al. (2025) evaluated it on two real-world EHR datasets: MIMIC-IV [7] and ODIR [11]. On MIMIC-IV, a binary classification setting is considered, while ODIR is used for a multi-label classification task.

4.1 Datasets and Tasks

We first introduce the freely accessible MIMIC-IV dataset, which comprises 418,100 patient admissions collected between 2008 and 2019 at the Beth Israel Deaconess Medical Center (BIDMC). The task under consideration involves five modalities from the dataset: demographics, laboratory test results, medications, diagnoses, and clinical notes. The data is split into training, validation, and test sets in an 8:1:1 ratio [7]. Following the CLEAR benchmark, we consider two tasks: **Critical Outcome Prediction**, which determines whether a patient will experience inpatient mortality or require transfer to the ICU within the next 12 hours, and **Hospitalization Prediction**, which predicts whether a patient presenting to the Emergency Department (ED) will be admitted to the hospital [15].

Table 2: Experimental results of critical outcome prediction on the MIMIC-IV dataset, comparing CLEAR against median and best performance across all baseline models for each metric.

Method	AUPRC	F1	Precision	Recall
Median (Baselines)	0.6702	0.5446	0.7460	0.4165
Best (Baselines)	0.7423	0.6484	0.7865	0.5733
CLEAR (Ours)	0.7844	0.6902	0.7399	0.6467

Baselines: MLP [20], M3Care [23], Dyhealth [25], UTDE [24], and HetMed [9].

Table 3: Experimental results of hospitalization prediction on the MIMIC-IV dataset, comparing CLEAR against median and best performance across all baseline models for each metric.

Method	AUPRC	F1	Precision	Recall
Median (Baselines)	0.9746	0.9071	0.9385	0.8620
Best (Baselines)	0.9858	0.9292	0.9607	0.9175
CLEAR (Ours)	0.9940	0.9579	0.9570	0.9588

Baselines: MLP [20], M3Care [23], Dyhealth [25], UTDE [24], and HetMed [9].

Table 4: Experimental results of ocular phenotype prediction on the ODIR dataset, comparing CLEAR against median and best performance across all baseline models for each metric.

Method	macro-AUC	micro-AUC
Median (Baselines)	0.8404	0.8963
Best (Baselines)	0.9010	0.9599
CLEAR (Ours)	0.9969	0.9986

Baselines: MLP [20], M3Care [23], Dyhealth [25], and HetMed [9].

The ODIR (Ocular Disease Intelligent Recognition) dataset is the second dataset used to evaluate CLEAR's performance. It contains demographic information (such as age and sex), bilateral color fundus photographs, and diagnostic keywords from 5,000 patients. The data is divided into training, validation, and test sets with a 7:1:2 split. The corresponding task is formulated as a multi-label classification problem over eight ocular diseases, referred to as **Ocular Phenotype Prediction** [11, 15].

4.2 Performance Evaluation

The experimental results in Tables 2, 3, and 4 demonstrate that CLEAR consistently outperforms all compared baseline methods ([20], [23], [25], [24], [9]) across both binary and multi-label classification settings.

On the MIMIC-IV dataset, CLEAR improves AUPRC by approximately 17%, resulting in a more reliable ranking of high-risk patients above low-risk patients. This directly reflects improved global discrimination between critical and non-critical cases. The F1 score increases by roughly 27%, showing a better balance between correctly identified critical patients (recall) and avoiding false alarms (precision). Most importantly, recall improves by more than 50%, indicating that CLEAR detects substantially more true critical cases, which is crucial for early warning scenarios such as ICU transfer or mortality prediction.

These gains directly indicate reduced representation contamination. The strong recall and improved overall ranking quality suggest that CLEAR learns more disentangled representations that better separate critical from non-critical patient states. A similar trend is observed in hospitalization prediction, where improvements across all metrics indicate more stable and reliable predictions across patients and conditions.

On the ODIR dataset, CLEAR achieves strong gains in macro- and micro-AUC, which measure how well the model separates each disease class (macro) and overall label prediction quality (micro). This shows that CLEAR reduces label confusion in highly correlated ocular diseases, where baseline methods often struggle due to overlapping visual and clinical patterns.

Overall, the results confirm that CLEAR effectively mitigates representation contamination by producing more discriminative and less entangled multimodal representations, leading to consistently better clinical decision performance [15].

5 DISCUSSION

The aforementioned performance improvements are particularly relevant in a clinical context. While precision remains comparable to baseline methods, recall increases substantially. Since missed detections of patient deterioration leads to serious consequences, high recall is often more important than marginal gains in precision. The results therefore suggest that CLEAR is better able to identify patients at risk while maintaining a similar false-positive rate.

Despite its strong performance, CLEAR does not model causal relationships in a strict sense. The counterfactual signals are generated from label-based prompts rather than real interventions or alternative patient outcomes. Therefore, the model mainly learns differences that help separate classes in the representation space, rather than true cause-and-effect relationships. This is important for interpretation, as the results should be understood as improving predictive separation rather than uncovering clinical causality. Additionally, the learned representations remain dependent on prompt design, as variations in phrasing may influence the resulting feature discrepancies and thus affect stability across tasks. Future work could investigate the integration of causal modeling approaches to improve robustness and interpretability.

6 CONCLUSION

This paper introduced CLEAR, a counterfactual learning approach for mitigating representation contamination in multimodal EHR data. Experiments on MIMIC-IV and ODIR show consistent performance gains, particularly in recall for critical outcome prediction, indicating improved robustness of learned representations. Overall, CLEAR demonstrates the effectiveness of counterfactual representation learning for multimodal clinical prediction tasks.

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A CLINICAL REFERENCE VALUES

This section provides detailed source information for the vital parameter values used in the discriminative and spurious feature illustrations (Figures 1 and 2). All values are derived from peer-reviewed clinical studies.

A.1 Normalization Ranges

All vital parameters were normalized to a [0,1] scale using the critical thresholds defined by the National Early Warning Score 2 (NEWS2) [12].

Table 5: Normalization ranges based on NEWS2

Parameter	Min	Max
Heart Rate (bpm)	40	131
Respiratory Rate (breaths/min)	8	25
Oxygen Saturation (%)	91	100
Systolic Blood Pressure (mmHg)	90	220
Body Temperature (°C)	35.0	39.1

A.2 Vital Parameters

The raw vital parameter values for each patient group before normalization are summarized below. ICU^+ values represent critical deterioration requiring ICU transfer [2]. ICU^- / $Fever^-$ values represent stable hospitalized adults [4]. $Fever^+$ values represent febrile patients without critical deterioration [1, 6].

Table 6: Summary of vital parameter values by group

Parameter	ICU$^+$	BASE	Fever$^+$
Heart Rate (bpm)	106	77	102
Respiratory Rate (breaths/min)	23	19	21
Oxygen Saturation (%)	97	98	95.2
Systolic Blood Pressure (mmHg)	111	135	127.8
Body Temperature (°C)	36.7	36.9	38.5

B ODIR REPRODUCTION

The following results present a reproduction of the experimental findings reported for the CLEAR model under the original evaluation setup. Each patient is annotated with binary labels for **eight disease categories** in a multi-label setting: Normal (N), Diabetes (D), Glaucoma (G), Cataract (C), Age-related Macular Degeneration (A), Hypertension (H), Myopia (M), and Other (O). Performance is evaluated using macro-AUC, micro-AUC, and per-label accuracy.

B.1 Input Modalities

Table 7: Available modalities and feature dimensions.

#	Modality	Dim	Description
1	Demographics	2	Age (norm.) and sex (bin.)
2	Fundus Images	1536	Left/right ResNet18 embeddings (768 each, concatenated)
3	Clinical Keywords	1536	Left/right ClinicalBERT [CLS] embeddings (768 each)

B.2 Performance Comparison

Table 8: Performance on the ODIR dataset. Results reported as mean ± std over 3 runs.

Label	Paper (AUC)	Reproduction (AUC)
N	0.9998	0.9999 ± 0.0000
D	0.9991	0.9994 ± 0.0003
G	0.9933	0.9956 ± 0.0024
C	0.9999	0.9975 ± 0.0013
A	0.9912	0.9989 ± 0.0001
H	0.9948	0.9983 ± 0.0006
M	0.9977	0.9830 ± 0.0150
O	0.9969	0.9977 ± 0.0004
Macro-AUC	**0.9969**	**0.9963 ± 0.0018**
Micro-AUC	**0.9986**	**0.9981 ± 0.0007**

The experimental setup was kept largely identical to the original configuration. Training hyperparameters, including the number of epochs, learning rate, optimizer choice, weight decay, batch size, embedding dimension, attention heads, number of transformer and cross layers, and the use of pretrained frozen image and text encoders, were all reproduced exactly as described in the original setup. Differences exist only in the hardware used and in the specific initialization of runs via different random seeds.

B.3 Source Code

https://github.com/jannikmenzel/CLEAR